CASE STUDY 6

RAILWAY FARE AND RESERVATION CALCULATOR

1. Problem Statement

Calculate fares for multiple passengers using distance slabs, travel classes, age-based concessions, reservation fees, and specified cancellation penalties.

2. Objectives

1. Read the number of passengers.
2. Calculate fares based on distance slabs.
3. Apply travel-class charges.
4. Provide age-based concessions.
5. Add reservation fees.
6. Calculate cancellation penalties.
7. Display each passenger's fare and the total amount.

3. Assumptions

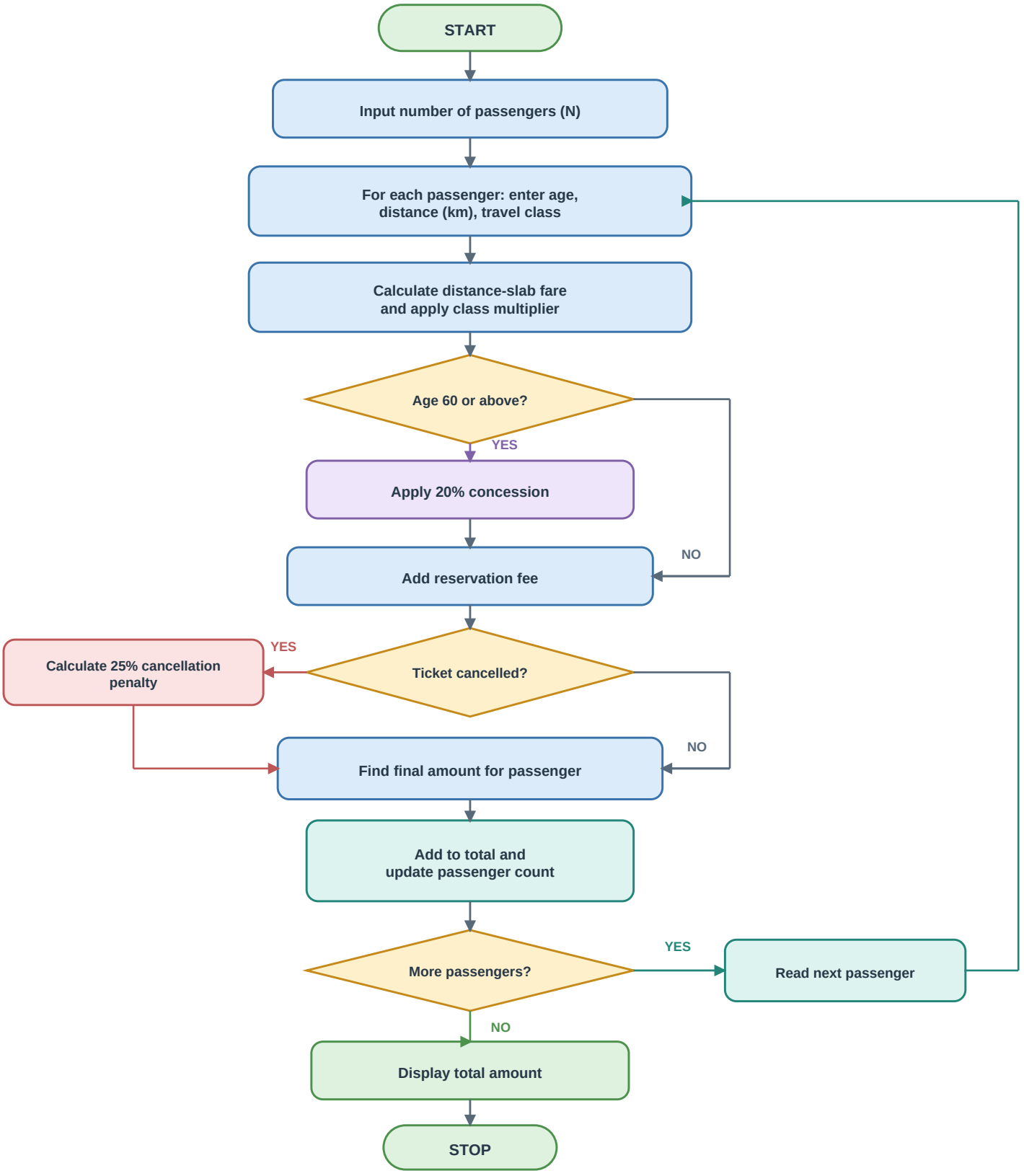
1. The user enters valid passenger details and a positive travel distance.
2. Sleeper class uses a multiplier of 1.0; AC class uses a multiplier of 2.0.
3. Distance slabs: first 100 km at 2.00 per km; next 400 km at 1.50 per km; distance above 500 km at 1.20 per km.
4. Passengers aged 60 or above receive a sample 20% concession on the class-adjusted fare.
5. Reservation fee is 50 per passenger.
6. If cancelled, the amount payable is a sample cancellation penalty of 25% of the fare after concession and reservation fee.
7. These numeric rules are sample assumptions for demonstrating the program. The supplied 100-case-study PDF does not provide exact rates or policy values.

4. Algorithm

1. Start the program.
2. Read the number of passengers and initialize the total amount to zero.
3. For each passenger, read age, distance, and travel class.
4. Calculate the basic fare using the distance slabs.
5. Apply the multiplier for the selected travel class.
6. If the passenger is aged 60 or above, apply the 20% concession.
7. Add the reservation fee.
8. Ask whether the ticket is cancelled. If yes, calculate 25% of the fare as the cancellation penalty; otherwise, use the ticket fare as the final amount.
9. Add the passenger's final amount to the running total and display it.
10. Repeat for all passengers.
11. Display the total amount and stop.

5. Flowchart

Coloured flowchart showing the passenger loop, age-concession decision, cancellation decision, and total calculation.



Flowchart key: Blue = process, yellow = decision, green = start/end, purple = concession step, red = cancellation-penalty step, teal = passenger/fee step.

6. Pseudocode

```
START
INPUT number_of_passengers
SET total_amount = 0

FOR each passenger
    INPUT age, distance, travel_class

    CALCULATE basic fare using distance slabs

    IF travel_class is Sleeper
        USE Sleeper multiplier
    ELSE IF travel_class is AC
        USE AC multiplier
    ELSE
        DISPLAY invalid class
        SKIP this passenger
    END IF

    IF age >= 60
        APPLY age concession
    END IF

    ADD reservation fee

    INPUT cancellation status
    IF ticket is cancelled
        CALCULATE cancellation penalty
        SET final amount = penalty
    ELSE
        SET final amount = ticket fare
    END IF

    ADD final amount to total_amount
    DISPLAY passenger amount
END FOR

DISPLAY total_amount
STOP
```

7. Complete Python Source Code

The code below follows the sample assumptions listed in Section 3.

```
total_amount = 0.0

n = int(input("Enter number of passengers: "))

for i in range(n):
    print("\\nPassenger", i + 1)

    age = int(input("Enter age: "))
    distance = float(input("Enter distance in km: "))

    print("1. Sleeper")
    print("2. AC")
    travel_class = int(input("Select travel class: "))

    # Calculate fare using distance slabs
    if distance <= 100:
        fare = distance * 2.0
    elif distance <= 500:
        fare = 100 * 2.0 + (distance - 100) * 1.5
    else:
        fare = 100 * 2.0 + 400 * 1.5 + (distance - 500) * 1.2

    # Apply travel class multiplier
    if travel_class == 1:
        fare = fare * 1.0
    elif travel_class == 2:
        fare = fare * 2.0
    else:
        print("Invalid travel class. Passenger skipped.")
        continue

    # Apply age-based concession
    if age >= 60:
        fare = fare * 0.80
```

```
    # Add reservation fee
    fare = fare + 50

    # Calculate amount payable
    cancel = input("Is ticket cancelled? (yes/no): ")

    if cancel.lower() == "yes":
        final_amount = fare * 0.25
        print("Cancellation penalty:", final_amount)
    else:
        final_amount = fare
        print("Ticket fare:", final_amount)

    total_amount = total_amount + final_amount

    print("Passenger amount:", round(final_amount, 2))

print("\\nTotal amount:", round(total_amount, 2))
```

8. Test Cases and Expected Output

Test Case 1: Senior passenger, Sleeper class, ticket not cancelled

Input values: age = 65, distance = 100 km, class = Sleeper, cancelled = no.

Calculation	Value
Basic fare: 100 × 2.00	200.00
Sleeper multiplier (× 1.0)	200.00
20% senior concession	–40.00
Fare after concession	160.00
Reservation fee	+50.00
Final amount	**210.00**

Expected output: Passenger amount = 210.00; Total amount = 210.00.

Test Case 2: Adult passenger, AC class, ticket cancelled

Input values: age = 35, distance = 150 km, class = AC, cancelled = yes.

Calculation	Value
First 100 km: 100 × 2.00	200.00
Next 50 km: 50 × 1.50	75.00
Basic fare	275.00
AC multiplier (× 2.0)	550.00
Reservation fee	+50.00
Fare before cancellation	600.00
Cancellation penalty: 25% × 600	150.00
Final amount payable	**150.00**

Expected output: Cancellation penalty = 150.00; Passenger amount = 150.00; Total amount = 150.00.

9. Conclusion

The program calculates railway fares for multiple passengers by applying distance-based slabs, travel-class multipliers, age concessions, reservation fees, and cancellation penalties. It displays the amount for each passenger and the final total.

Important note: The original assignment statement names the required features but does not specify exact fare slabs, concession percentages, reservation fees, or cancellation rules. The numerical values in this document are clearly marked sample assumptions; replace them with your faculty's prescribed values if provided.